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An empirical study of personalized advertising recommendation based on DBSCAN clustering of sina weibo user-generated content

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Abstract

Today, with the dividend of mobile Internet traffic disappearing, mobile advertising is focusing on powerful algorithms of personalized recommendation. Sina Weibo has abundant advertising spaces and the accuracy of its recommendation directly affects user experience. UGC (user-generated content) is the selected or even original edited content for interaction through users' subjective interests, which reflects users' inner interests within a certain period of time. Adaptive DBSCAN clustering is performed on Weibo UGC, and the clustering center represents the user's interest to a certain extent. By refining the text theme of advertisements to be recommended, the similarity between clustering center and the recommended theme is obtained, and the advertising category with high similarity is recommended to user. The experiment proves that the method is effective.

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Keywords: Personalized advertising recommendation; adaptive DBSCAN clustering; user interest; Sina Weibo.

1. Introduction

1.1. Research background

Sina Weibo is a popular social network platform in China, on which users can create, obtain and spread information in real time based on their social relations. As an important part of Sina Weibo (hereinafter referred to as Weibo) revenue, advertising has an irreplaceable marketing value. high-quality news feed ads recommended

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based on personal information can significantly improve the advertising effect. how to mine user's interests according to user's information and distribute ads matching user's interest is important.

The development of social platform advertising has gone through three stages: (1) social platforms are only used as communication channel; (2) interactive feedback is added to form the economy and culture of fans; (3) through technical means, advertising can achieve accurate delivery and reach users effectively. Among them, how to excavate, identify, collect and predict user interest is the foundation of realizing accurate marketing. This paper uses the user data of social platforms to mine user interests and achieve accurate advertising. Specifically, we implement UGC and DBSCAN clustering to extract users' short-term interests, distribute personalized advertisements. Experiment proves the feasibility of the proposed method based on UGC. Based on the traditional empirical method to determine DBSCAN clustering parameters, we verify that the adaptive method has obvious timeliness and superiority.

1.2. Literature review

J. Bobadilla et al. believe that the internal functions of the recommendation system are mainly represented by the filtering algorithms which are classified into collaborative filtering, population filtering, content-based filtering, and hybrid filtering [1]. One of the common tasks of recommendation systems is to improve the user experience through personalized recommendations based on prior implicit feedback [2]. Unlike explicit feedback, which has been studied more extensively, implicit feedback does not directly input their preferences to the user.

Billsus D et al. present a framework for adaptive news access based on specific machine learning techniques. Machine learning algorithms are used to predict user models based on implicit and explicit feedback from users on mobile terminals [3]. Hanyu Li et al. believe that building user interest models faces three major challenges, namely, information acquisition of initial users, user information representation and description of user interest changes [4]. Collaborative filtering is a widely adopted recommended method, but this method faces the problems of data sparsity and cold start. To overcome the problems, Maryam Khanian Najafabadi et al apply the users' implicit interaction records with items to efficiently process massive data by employing association rules mining [5].

Chinese social platforms are also making new breakthroughs in personalized advertising recommendation. Yang Bo et al. proposed a collaborative filtering recommendation method to improve the performance of collaborative filtering recommendation [6]. The personalized product recommendation method proposed by He Feng and others enters data from user images and other UGC content, including photo tags with user comments and comments from others on social networking sites [7].

Research on personalized advertising recommendation systems at home and abroad includes multiple platforms and entry points, but the content is relatively loose. Weibo is characterized by the fact that user data is concentrated in a large text set, which contains users' distinct interest characteristics. Different from the above studies, this paper makes full use of UGC data to advertising recommendation, and adds an adaptive method to determine parameters in the interest extraction stage to improve the timeliness of recommendations.

2. Design of research methods

The experimental process and technical route are shown in Fig. 1. The research mainly includes three parts:

- Extraction of user interest: The original and forwarded microblogs of the user are crawled and the text is preprocessed and clustered. The text set obtained can represent the user's short-term interest.
- Calculation of similarity between user interest and advertising text: The clustering results are converted into vectors as features of user interest, and keywords are extracted as advertising themes. Similarity calculation is conducted between themes and cluster centers and ads with high similarity are determined as push categories.
- Determine the category of recommended advertisements: We can obtain the advertising categories that users are interested in through similarity. Users' feedback on the pushed ads will prove the recommendation effect.

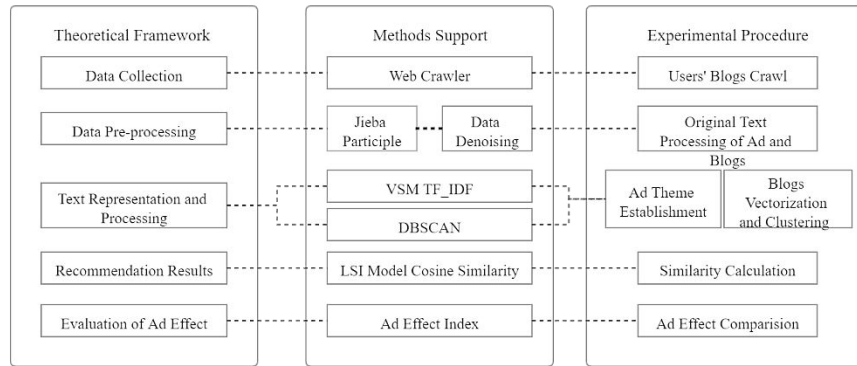


Fig. 1. Technology roadmap.

2.1. The type of ads studied

The type of ads studied is news feed ads whose presentation form is the same as microblog, but the content is advertising promotion. This kind of advertisement has 'AD' tag in the upper left corner for users to identify. The ads also contain interactive features, such as forwarding, like, and comment. The users' interactions with the ads (forward, like and comment) verify whether the users are interested in the ads (or the categories of the ads). The paper will select the user's likes status of the ad to determine whether the user is interested in the ad.

2.2. The acquisition and processing of microblog posts

The acquisition and processing process of microblog includes: preprocessing of original text; text feature extraction; DBSCAN clustering [8]. Text preprocessing is the process of converting text information from human language to machine readable format for subsequent processing. Users' blogs contain lots of special characters that are useless for text analysis, such as punctuation marks, special symbols and repeated meaningless phrases. Jieba word segmentation is a Python Chinese phrase segmentation, which can be used for word segmentation, part-of-speech tagging, keyword extraction and other functions of Chinese text. Then the word frequency features of TF-IDF were used to extract text features. By using the empirical method to adjust the weight, the ideal clustering effect is obtained. Table 1 shows part of the weight coefficients.

Table 1. Weight of each part of speech.

Speech code	Speech	Weight
n	Noun	3.0
nr	Person's name	3.5
ns	Place names	3.5
nt	Organization	4
v	The verb	2
a	Adjectives	1

2.3. The theme of Weibo ads

Using TF-IDF model to calculate the similarity, the short text similarity is very low. In the case of less corpus, the similarity with the text of the same topic is not high. Therefore, this paper adopts LSI model to extract advertising text themes. Each topic contains all known words, but they have different probability distributions in different topics. In this paper, 10 ads in each of 48 fields are selected. Assuming that each ad text has 480 themes,

the text vector can be reduced to 480 dimensions through the LSI model, and each component represents the weight of the corresponding theme. Finally, the similarity between advertising text and user interest cluster is obtained through similarity queries.

3. The empirical research

This experiment was carried out under the Window10 system, with Intel(R) Core(TM) I5-7200U CPU @2.50ghz 2.70ghz processor and 4GB memory. The experimental code is Python. Running software are Spyder and Jupyter Notebook. The data visualization tool is Tableau 10.5.

3.1. Experimental text collection

50 Sina Weibo users were randomly selected for the experiment, and the number of their published microblog posts (including forwarding) was shown in Fig. 2. (a).

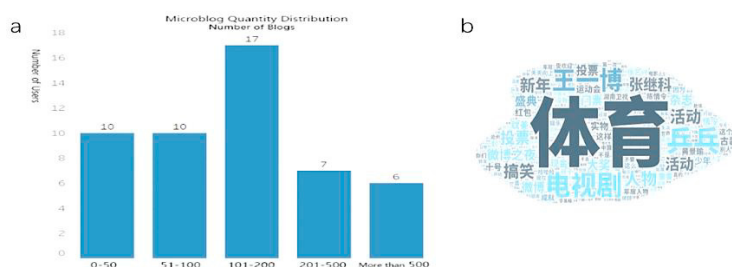


Fig. 2. (a) microblog quantity distribution of experimental users; (b) word cloud display of clustering results of a user.

3.2. Clustering results

All the original and forwarded microblogs of 50 users were selected. The cluster center of interest of each user was obtained by DBSCAN clustering algorithm. The parameters (ϵ , MinPts) are used to describe the density of the neighborhood sample distribution. ϵ describes the neighborhood distance threshold for a sample, and MinPts describes the threshold for the number of samples in the neighborhood. The experiment compares the two methods of parameter determination: empirical method and adaptive method [9]. Fig. 2. (b) shows the clustering results of a user's blog posts.

3.3. The theme modeling of microblog advertisements

The 480 ads in the experimental sample cover 48 official areas of Weibo. Table 2 shows the classification details.

Users with at least 5 clustering centers after interest clustering were selected for comparison experiment. A total of 10 users were selected to calculate the similarity between the user's interest center and the advertisements in 48 areas. Among them, the average value of the similarity of 10 advertisements in a certain field is the similarity between the user's interest and the field. For example, the average similarity of A user's 10 advertisements in the field of technology is 0.5, then 0.5 represents the interest of A user in the field of technology.

Table 3 shows DBSCAN algorithm parameters determined by empirical and adaptive algorithm respectively. After obtaining the two clustering centers, 10 users' interests in advertisements in the two fields of 'international affairs, science' were calculated.

Table 2. The field of Weibo advertising.

international affairs	science	economics	celebrity	law	wedding
sports	health	regimens	history	antique	dance
positive energy	game	parenting	food	TV series	variety show
design	fashion	anime	pet	photography	model
social	technology	digital	stock market	household	housing property
car	sports fitness	slimming	military	music	movie
dispel rumors	government affairs	tourism	campus	joke	emotion
agriculture	art	makeup	religious	book	constellation

Table 3. Users' interest in advertising in the two areas.

Users	Parameter determination method	Interests	
		International affairs	Science
User1	Empirical	0.376	0.405
	Adaptive	0.376	0.405
User2	Empirical	0.146	0.136
	Adaptive	0.146	0.136
User3	Empirical	0.413	0.366
	Adaptive	0.413	0.366
User4	Empirical	0.242	0.261
	Adaptive	0.242	0.261
User5	Empirical	0.307	0.205
	Adaptive	0.307	0.205
User6	Empirical	0.323	0.334
	Adaptive	0.323	0.334
User7	Empirical	0.296	0.465
	Adaptive	0.296	0.465
User8	Empirical	0.158	0.117
	Adaptive	0.158	0.117
User9	Empirical	0.15	0.492
	Adaptive	0.15	0.492
User10	Empirical	0.164	0.112
	Adaptive	0.164	0.112

3.4. Recommendation threshold setting

Ad recommendation threshold is the similarity threshold for determining whether to recommend ads in this area. If the similarity is higher than the threshold, it is deemed to recommend advertisements in this field to the user. Fig.3 compares the number of ad categories that the above 10 users are interested in under different thresholds. When the threshold is 0.04, the number of advertisements that match is appropriate. In this experiment, 0.04 is selected as the advertising recommendation threshold.

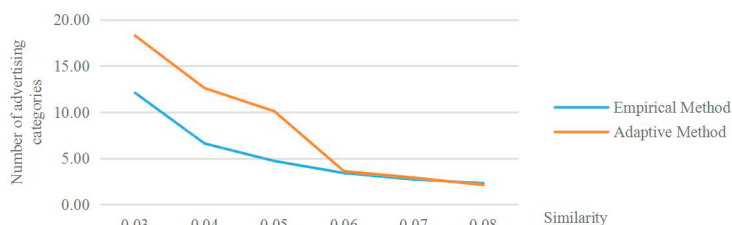


Fig. 3. Influence of thresholds on the number of recommendations.

3.5. Category accuracy verification of recommended ads

When the similarity is more than 0.04, the category of ads can be recommended. According to the threshold, the interested ad categories of 10 users were obtained, and then verified by comparing the liked ad categories of users. The literature defines three measurement indexes that are widely used in the quality of recommendation system: (1) Accuracy: the proportion of the number of interested advertisements recommended to users in the total number of recommendations; (2) Recall rate: the proportion of the number of interested advertisements recommended to users in the total number of interested advertisements of users; (3) F1: The combination of the first two indicators.

The experiment randomly selected 10 users from the data set for the experiment and obtained the accuracy.

Table 4. Accuracy of advertising recommendation category.

Users	Parameter determination method	Recommend number of success	Total recommended quantity	Precision rate	Recall rate	F1
User1	Empirical	3	11	0.27	0.50	0.35
	Adaptive	4	18	0.22	0.67	0.33
User2	Empirical	5	7	0.71	0.38	0.50
	Adaptive	9	12	0.75	0.69	0.72
User3	Empirical	2	8	0.25	0.50	0.33
	Adaptive	4	14	0.29	1.00	0.44
User4	Empirical	2	3	0.67	0.40	0.50
	Adaptive	3	3	1.00	0.60	0.75
User5	Empirical	2	5	0.40	0.33	0.36
	Adaptive	4	7	0.57	0.67	0.62
User6	Empirical	5	6	0.83	0.56	0.67
	Adaptive	5	6	0.83	0.56	0.67
User7	Empirical	2	4	0.50	1.00	0.67
	Adaptive	2	3	0.67	1.00	0.80
User8	Empirical	2	6	0.33	0.33	0.33
	Adaptive	6	48	0.13	1.00	0.22
User9	Empirical	5	8	0.63	0.71	0.67
	Adaptive	4	6	0.67	0.57	0.62
User10	Empirical	5	8	0.63	1.00	0.77
	Adaptive	5	9	0.56	1.00	0.71

The experimental results are shown in Table 4. Fig.4 compare the precision rate, recall rate and F1 value of the recommendation effect of 10 users' advertisements. The accuracy of the recommended method for determining parameters by empirical method is between 0.2 and 0.8, the recall rate is between 0.4 and 1, and F1 is between 0.3 and 0.7. The accuracy of the ad recommendation method based on adaptive parameter determination is between 0.2 and 1, the recall rate is between 0.6 and 1, and F1 is between 0.4 and 0.8. It can be seen that the adaptive clustering parameter determination method is more efficient than the empirical method, with higher accuracy, recall rate and F1 value. With the increase of user data and the increase of advertising text, the superiority of the adaptive parameter determination clustering method is more obvious.

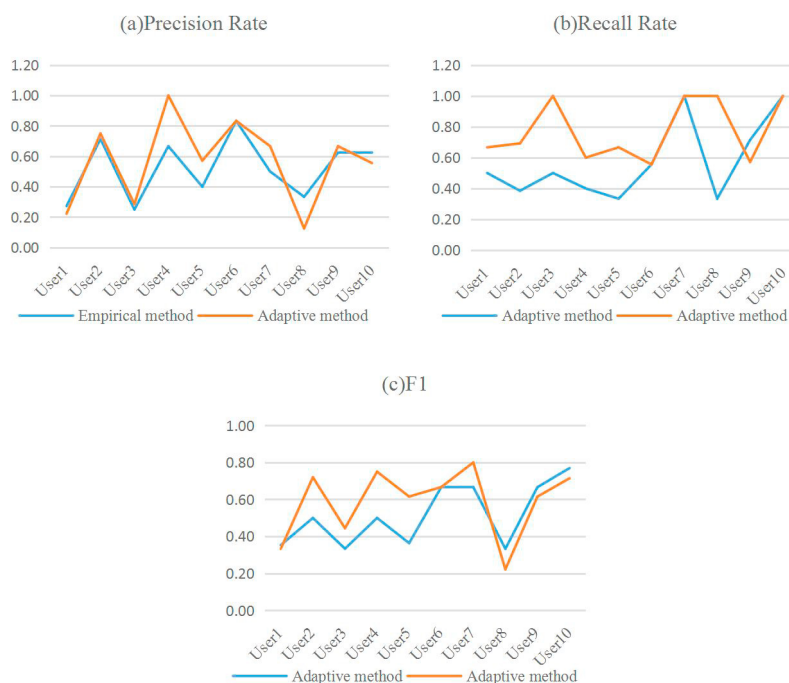


Fig. 4. Recommend effect under two parameter determination methods: (a) precision rate; (b) recall rate; (c) F1 values.

4. Conclusion

The paper analyzes the current situation of news feed ads on Weibo, aiming at the hot topic of personalized advertising recommendation. The experiment processed and clustered UGC text, mined user interest, extracted ads text theme by LSI model, and calculated the similarity between ads text and user interest. The filtered ads text is used as the recommendation set, and then verified the accuracy of the recommendation. Experiment shows that DBSCAN clustering method is effective in describing topics that users are interested in. Moreover, compared with empirical method, adaptive determination parameter has higher accuracy and recall rate in screening ads related to users' interest. There is still room for improvement: users' customized tags can be mined to extract long-term interest, and users with high similarity can be mined to get potential interests. The interaction mechanism of microblog includes the expression of attitude, which is valuable for mining users' negative interests.

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